

ATA BERK ÖZTÜRK

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[LinkedIn](#) | [GitHub](#)

Ankara - Türkiye

ABOUT

- Physics Engineer specializing in computational condensed matter physics, DFT workflows, and high-performance computing.
- Interested in early-career R&D roles in scientific computing, defense technologies, and materials modeling.

EDUCATION

Ankara University

B.Sc. in Physics Engineering (English)

GPA: 3.13/4.00

2022 – 2026

Ankara, Türkiye

Ankara University

Master's in Physics Engineering

2026 – Expected 2028

Ankara, Türkiye

EXPERIENCE

Computational Condensed Matter Physics Group

Researcher

February 2024 – Present

Ankara University

- Performed high-performance DFT simulations to analyze electronic and magnetic properties of 2D materials, contributing to data-driven material discovery.

EDULINE IT

Engineering Intern

July 2025 – August 2025

İstanbul

- Optimized engineering workflows on Rocky Linux by performing AxB node benchmarking and configuring custom-compiled software environments, utilizing Python (NumPy, Pandas) for automated data analysis.

Ankara University Institute of Nuclear Sciences

Intern

August 2025 - September 2025

- Configured and compiled nuclear distributions using Geant4 and ROOT frameworks in a Linux environment for precision data modeling.

PROJECTS

TÜBİTAK 2209-A

Undergraduate Researcher

2025 - 2026

- Analyzing DFT-based simulations on surface adsorption and band structure modifications of 2D materials.

TEKNOFEST - International UAV Competition

Member of ORION

2022 - 2023

- Contributed to simulation and calculation modules for UAV systems as part of the ORION team.

UNDERGRADUATE THESIS

Researcher

- Developed Gradient Boosting and Random Forest models achieving 95%+ accuracy in predicting magnetic material properties using VASP-derived electronic structure data; constructed a dataset of 2,036 materials from 316k data points.

TECHNICAL SKILLS

HPC & Infrastructure

- Linux, Slurm, OpenHPC, Docker, Git, Bash

Computational Physics

- VASP, Quantum ESPRESSO, Geant4, ROOT, Qiskit

Engineering Tools

- ANSYS (Zemax, OpticStudio, SpaceClaim, Maxwell)

Programming & Data

- Python (Pandas, NumPy), C++, MATLAB, LaTeX

Languages: Turkish (native), English (B2+)

CERTIFICATIONS & REFERENCES

CERTIFICATIONS: QBronze - Qiskit, NUMESYS Optical Camp Success Certificate (ANSYS Zemax, Speos), ARCHER2 HPC Driving License (High Performance Computing), Defense Industry Academy - Career & Competency Programs

REFERENCES: Upon request